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Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 7 – Administration Systeme

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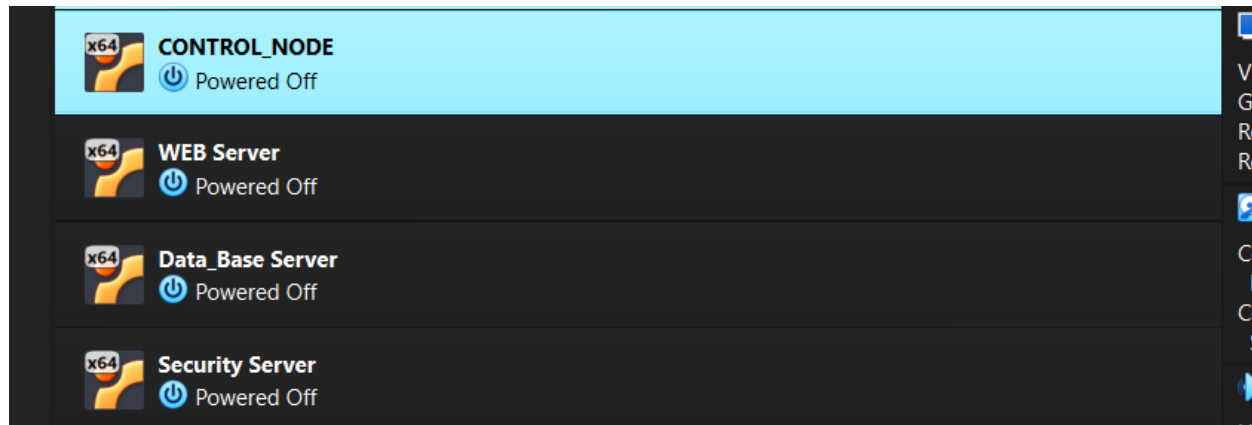
Niveau : L3

Le 19 avril 2026

L'objectif de ce TD est de :

1. installer Ansible
2. créer un inventaire
3. structurer un projet avec des rôles
4. utiliser des templates Jinja2
5. lancer un playbook complet

Les Serveurs



- Installation d'Ansible sur CONTROLE NODE

```
donsam@donsam:~$ sudo apt update
[sudo] password for donsam:
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Réception de :3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Réception de :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Réception de :5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1 608 kB]
Réception de :6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1 901 kB]
Réception de :7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [257 kB]
Réception de :8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21,5 kB]
Réception de :9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10,8 kB]
Réception de :10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2 822 kB]
Réception de :11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [345 kB]
Réception de :12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Réception de :13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16,9 kB]
Réception de :14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2 957 kB]
Réception de :15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [658 kB]
Réception de :16 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [689 kB]
Réception de :17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [288 B]
Réception de :18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1 169 kB]
Réception de :19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Réception de :20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1 666 kB]
```

```
donsam@donsam:~$ sudo apt install ansible -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python3-ntlm-auth
  python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xmltodict
Paquets suggérés :
  cowsay sshpass python3-trio python3-aiioqueue python3-h2 python3-httpx python3-httpcore python-lockfile-doc
Les NOUVEAUX paquets suivants seront installés :
  ansible ansible-core python3-argcomplete python3-dnspython python3-kerberos python3-libcloud python3-lockfile python3-requests-ntlm
  python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xmltodict
0 mis à jour, 15 nouvellement installés, 0 à enlever et 32 non mis à jour.
Il est nécessaire de prendre 19,5 Mo dans les archives.
Après cette opération, 315 Mo d'espace disque supplémentaires seront utilisés.
```

```
donsam@donsam:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/donsam/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/donsam/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

- Configuration SSH sur les machines virtuelles

```
donsam@donsam:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/donsam/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/donsam/.ssh/id_rsa
Your public key has been saved in /home/donsam/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:stnWJDRO9//ti5gbtBFpn3M+z9pHGdvbbIQsusGP1lw donsam@donsam
The key's randomart image is:
+---[RSA 3072]-----+
|
|      .      .
|    o +  +
|   . o o. o ..
|   . S oo = o=
|   = =..+ Eo+
|   o o +=.+ *o
|   . .+B..=B
|   o*o.o=o
+---[SHA256]-----+
donsam@donsam:~$
```

```
donsam@donsam:~$ ssh-copy-id -p 2222 web1@192.168.56.106
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/donsam/.ssh/id_rsa.pub"
The authenticity of host '[192.168.56.106]:2222 ([192.168.56.106]:2222)' can't be established.
ED25519 key fingerprint is SHA256:nvAXgPXmikXScEtAYT2IKb0tL230yzCsAE6iJw9FrFw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
web1@192.168.56.106's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh -p 2222 'web1@192.168.56.106'"
and check to make sure that only the key(s) you wanted were added.
```

```
donsam@donsam:~$ ssh-copy-id -p 2222 web2@192.168.56.107
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/donsam/.ssh/id_rsa.pub"
The authenticity of host '[192.168.56.107]:2222 ([192.168.56.107]:2222)' can't be established.
ED25519 key fingerprint is SHA256:8WMMWbZ4uWNb40Vldi8EHMMxD82BUGCXJ2CrHBLk7VRg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
web2@192.168.56.107's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh -p 2222 'web2@192.168.56.107'"
and check to make sure that only the key(s) you wanted were added.
donsam@donsam:~$
```

```
donsam@donsam:~$ ssh-copy-id security@192.168.56.103
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/donsam/.ssh/id_rsa.pub"
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:CrQkkb8FulZ0NMQSZE6H9DYtPWXhboGJwUxG1zBKK4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
security@192.168.56.103's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'security@192.168.56.103'"
and check to make sure that only the key(s) you wanted were added.

donsam@donsam:~$ _
```

- Connexion entre les serveurs établies

```
donsam@donsam:~$ ssh web1@192.168.56.106
ssh: connect to host 192.168.56.106 port 22: Connection refused
donsam@donsam:~$ ssh -p 2222 web1@192.168.56.106
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Apr 20 01:36:34 AM UTC 2026

System load:          0.0
Usage of /:            12.0% of 24.44GB
Memory usage:         10%
Swap usage:           0%
Processes:            112
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe2c:5f41

Expanded Security Maintenance for Applications is not enabled.

42 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Apr 19 18:34:02 2026 from 192.168.1.29
```

```
donsam@donsam:~$ ssh -p 2222 web2@192.168.56.107
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Apr 20 01:37:37 AM UTC 2026

System load:            0.18
Usage of /:             11.7% of 24.44GB
Memory usage:          5%
Swap usage:            0%
Processes:             128
Users logged in:       1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea2:3e1b

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

42 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Apr 19 18:40:46 2026 from 192.168.1.128
web2@WEB2:~$
```

```
donsam@donsam:~$ ssh security@192.168.56.103
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of lun. 20 avril 2026 01:38:18 UTC

System load:            0.11
Usage of /:             43.7% of 11.21GB
Memory usage:          7%
Swap usage:            0%
Processes:             120
Users logged in:       1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea1:179f

La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

security@security:~$
```

Création du projet Ansible

```
donsam@donsam:~$ mkdir tp-ansible
donsam@donsam:~$ cd tp-ansible
donsam@donsam:~/tp-ansible$
```

- Fichier inventaire

```
GNU nano 7.2                                inventory.ini *
[web1]
192.168.56.106

[web2]
192.168.56.107

[security]
192.168.56.103
```

- Test de connectivité

```
donsam@donsam:~/tp-ansible$ ansible all -i inventory.ini -m ping
192.168.56.103 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
192.168.56.106 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
192.168.56.107 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
donsam@donsam:~/tp-ansible$
```

- Création des rôles

```
donsam@donsam:~/tp-ansible$ ansible-galaxy init roles/web1
- Role roles/web1 was created successfully
donsam@donsam:~/tp-ansible$ ansible-galaxy init roles/web2
- Role roles/web2 was created successfully
donsam@donsam:~/tp-ansible$ ansible-galaxy init roles/security
- Role roles/security was created successfully
donsam@donsam:~/tp-ansible$ ls roles
security web1 web2
donsam@donsam:~/tp-ansible$
```

- Rôle du WEB

```
# tasks file for roles/web2
- name: Installer Nginx
  apt:
    name: nginx
    state: present
    update_cache: yes

- name: Demarrer Nginx
  service:
    name: nginx
    state: started
    enable: yes

- name: Deployer la page d'accueil
  template:
    src: index.html.j2
    dest: /var/www/html/index.html
```

- Rôle du Base de données

```
GNU nano 7.2 roles/web2/tasks/main.yml
---
# tasks file for roles/web2
- name: Installer MySQL
  apt:
    name: mysql-server
    state: present
    update_cache: yes
```


- Rôle du Sécurité

```
GNU nano 7.2 roles/security/tasks/main.yml
---
# tasks file for roles/security
- name: Installer UFW
  apt:
    name: ufw
    state: present

- name: Autoriser SSH
  ufw:
    rule: allow
    port: 22

- name: Activer le firewall
  ufw:
    state: enable
```

- Création du template Jinja2

```
GNU nano 7.2 roles/web1/templates/index.html.j2 *
<h1>Bienvenue sur {{ inventory_hostname }}</h1>
<p>Serveur deploye automatiquement avec Ansible</p>
```

- Création du playbook principal

```
donsam@donsam:~/tp-ansible$ sudo nano site.yml_
```

```
GNU nano 7.2                                site.yml *
- hosts: web1
  become: yes
  roles:
    - web1

- hosts: web2
  become: yes
  roles:
    - web2

- hosts: security
  become: yes
  roles:
    - security
```

- Vérification de l'utilisation de mot de passe

```
Last login: Mon Apr 20 01:45:28 CEST 2020 from 192.168.56.106
web1@WEB1:~$ sudo visudo
[sudo] password for web1: _

Last login: Mon Apr 20 01:45:30 2020 from 192.168.56.106
web2@WEB2:~$ sudo visudo
[sudo] password for web2: _

security@security:~$ sudo visudo
[sudo] password for security: _
```

- Exécution du playbook

```
donsam@donsam:~/tp-ansible$ ansible-playbook -i inventory.ini site.yml

PLAY [web1] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

TASK [web1 : Installer Nginx] *****
```

1. Pourquoi utiliser Ansible ?

Ansible permet d'**automatiser** la configuration et la gestion de plusieurs serveurs à la fois. Au lieu de se connecter manuellement sur chaque serveur pour faire des modifications, on écrit un

playbook une seule fois et Ansible l'exécute sur tous les serveurs simultanément. C'est rapide, reproductible et sans erreur humaine.

2. Quel est le rôle du fichier inventory ?

Le fichier `inventory.ini` liste **tous les serveurs** que Ansible doit gérer, organisés en groupes. Il indique à Ansible **où se connecter** (adresse IP), **avec quel utilisateur** et **sur quel port**. C'est l'annuaire de votre infrastructure.

3. Quelle différence entre copy et template ?

- **copy** : copie un fichier **statique** tel quel sur le serveur distant. Le contenu ne change pas.
- **template** : copie un fichier **dynamique** en remplaçant des variables par leurs valeurs. Par exemple, mettre automatiquement le bon nom de serveur ou l'adresse IP selon la machine cible.

4. Pourquoi séparer les rôles ?

Séparer les rôles permet de :

- **Organiser** le code en modules indépendants (web, db, security)
- **Réutiliser** un rôle sur plusieurs projets
- **Maintenir** facilement — si Nginx change, on modifie uniquement le rôle web
- **Collaborer** en équipe sans conflits — chacun travaille sur son rôle

Conclusion

Ce TP nous a permis de découvrir Ansible comme outil d'automatisation dans un environnement multi-serveurs virtualisé. Grâce aux concepts de playbook, d'inventory et de rôles, nous avons pu configurer et gérer plusieurs serveurs de façon organisée et efficace, en évitant les interventions manuelles répétitives.